

AI Education from the Educator's Perspective: Best Practices for an Inclusive AI Curriculum for Middle School

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ABSTRACT

Artificial Intelligence (AI) and its applications have a strong impact on people's lives worldwide. Therefore, an increasing number of professionals and academics are focusing on the development of AI education programs for K-12. Part of these programs is developed for formal educational settings, such as school classrooms, which are complex systems with several variables interwoven. The current study concentrates on the teachers' perspective on the implementation of an AI education curriculum for middle schools located in disadvantaged areas of Europe. Given these premises, a particular focus is devoted to how AI education can foster inclusion and be more accessible. Feedback was gathered through 582 logbooks provided by educators, and an international focus group conducted at the end of the curriculum implementation. This information helped to understand educators' points of view on the AI curriculum, investigating how these topics can stimulate reflections around inclusion and diversity.

CCS CONCEPTS

• **Applied computing** → **Education**; • **Human-centered computing**;

KEYWORDS

AI Education, Digital Competencies, Inclusive Education

ACM Reference Format:

Giorgia Gibellini, Valeria Fabretti, and Gianluca Schiavo. 2023. AI Education from the Educator's Perspective: Best Practices for an Inclusive AI Curriculum for Middle School. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3585747>

1 AI EDUCATION IN FORMAL EDUCATIONAL CONTEXTS

The need for Artificial Intelligence (AI) literacy in K-12 education programs is becoming increasingly important as technology continues to advance and play a significant role in our daily lives. AI is already pervasive across many everyday tasks and is expected

to continue to shape the future of work, society and education. Therefore, providing students with an understanding of AI and its capabilities is crucial for preparing them for the future [32].

One of the key aspects of AI literacy for K-12 students is understanding the basics of how AI works and its potential applications. This includes understanding concepts such as machine learning, natural language processing, and computer vision. However, AI literacy also includes understanding the ethical and societal implications of AI and critically considering the potential limitations and biases of AI systems [17, 20, 30].

AI literacy is considered crucial for the development of children for two main reasons: it empowers them and makes them safer in technological environments [27]. The need to look further than the existing digital competencies, towards digital competencies for AI, is also recognized by international organizations such as UNESCO [26] and UNICEF [27, 28].

Various programs and digital tools have been developed primarily in English to support AI education activities [8, 25, 32]. Among them is the AIK12 initiative, which is dedicated to providing K-12 teachers with the resources and training they require to incorporate AI into their classrooms [8, 25]. Although several studies have examined the implementation of AI education in informal and non-formal educational contexts, such as after-school programs, museums [16], and family settings [6, 19], effectively integrating AI education into formal education settings remains a challenge. Formal education is the primary mode of teaching in compulsory education, making it critical to find effective solutions for integrating AI education into this setting. However, only a few reported initiatives have been created in collaboration with educators, and many are not easily accessible, particularly for teachers without an ICT background [15, 21]. Therefore, actively engaging teachers in the development of AI educational programs remains an open challenge.

1.1 Pedagogical frameworks

In K-12 formal education, various pedagogical frameworks have been adopted to develop interventions for AI education. One such approach involves integrating AI concepts and applications into existing curricula. For instance, machine learning can be integrated into math classes, while natural language processing can be incorporated into language arts classes. This approach allows students to learn about AI in the context of subjects they are already familiar with, making the material more relatable and engaging.

Another approach is to create dedicated courses that focus solely on AI technology and its applications. These courses may cover topics such as the history of AI, the basics of machine learning, and

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585747>

the ethical and societal implications of the technology. By offering a dedicated course on AI, educators can ensure that students receive a comprehensive and in-depth understanding of the topic[2].

Different pedagogical strategies can be employed to engage students in learning about AI and developing their skills. Project-based learning, for instance, has proven to be an effective approach for increasing student engagement [30, 32]. Through project-based learning, students work on real-world AI projects, such as building a chatbot or creating a machine-learning model to predict a certain outcome. This hands-on approach enables pupils to learn by doing and apply the concepts they have learned.

Another effective approach is problem-based learning, in which students are presented with a real-world problem and are required to use AI techniques to solve it [13]. This approach not only teaches students how to use AI to solve real-world problems but also enhances their critical thinking and problem-solving abilities.

Lastly, incorporating ethics and societal implications of AI in the curriculum is also important [30]. As AI is becoming more prevalent in our lives, it is important for students to understand the potential impact of AI on society and to be equipped with the knowledge to make informed decisions about the use of AI. This can include discussions about the potential biases and limitations of AI, as well as the ethical considerations surrounding the use of AI.

When considering pedagogical frameworks for AI education two main factors should be taken into account: the specificity of the topic and the fast development of AI. Many educators, in fact, struggle to keep up with the technological advancements in the pedagogical field [31] resulting in a lack of confidence when coming to those topics and a need for a clear and well-structured framework, materials, and activity planning. Another important aspect is how to support schools and teachers to make their AI education classrooms more inclusive and accessible.

2 DIVERSITY AND INCLUSION IN AI EDUCATION

Diversity in Education refers to the differences between pupils in terms of gender, social skills, ethnic background, interests, linguistic skills, and all the possible shades of diversity between individuals [1]. Optimal learning environments need to take into account those differences, identify the one creating educational implications and shape the learning activities to respond to the needs of a diverse audience [1].

Given the increasing diversity in educational settings, ensuring inclusive education has become a crucial element in promoting equity in society's future. This topic is gathering more and more interest and has been defined by UNICEF as following "a dynamic process that is constantly evolving according to the local culture and context, [...] to celebrate diversity, promote participation and overcome barriers to learning and participation for all people" [27].

Narrowing the discussion around diversity and inclusion in K-12 education in the field of Computer Science, the literature presents many best practices for ensuring inclusive computing teaching embedding, for instance, conducting retrospective work on materials and programs developed taking into consideration the diversity of students; adopting a prospective approach for curricular materials; or include professional development on the inclusion in

teachers path [11]. Research on technology and inclusive education has demonstrated that technology has the potential to enhance students' engagement and comprehension. However, it is important to note that the positive impact of technology in the classroom is dependent on how it is used [22].

In the present work, we explore the potentialities of inclusive AI literacy, exploring the importance of teacher perspectives in the adoption of AI curricula. We report on the analysis of 582 logbooks and an international focus group with middle school teachers that have implemented AI educational activities in their classes.

3 METHOD

This study follows the implementation of an educational intervention targeting AI education that involved 114 middle school teachers (from both STEM and non-STEM subjects) and 1063 students. Schools participating in the project were all located in disadvantaged urban or rural areas of four different European countries (Italy, Bulgaria, Greece and Romania). The AI curriculum was developed adopting an interdisciplinary and project-based approach and it consists of four different learning modules for a total of 32 hours of in-class lessons. The four modules cover: Module 1. Introduction to AI (introductory concepts and AI-related disciplines), Module 2. How AI work (algorithms and machine learning), Module 3. Ethics and AI (societal impact and ethical considerations for AI), and Module 4. Life on the Net (the impact of AI in social media). Before implementing the activities in class, all the teachers underwent 16 hours of training on the modules and related activities from professional teacher trainers. The assessment of the AI curricula has been conducted through two main research activities: the collection and analysis of teacher logbooks (post-lecture surveys) and an international focus group with 12 teachers.

3.1 Logbooks

The logbooks consist of an online survey filled out by educators at the end of each lecture. The objective of the logbooks is to gather quantitative data regarding the lecture procedure and at the same time receive qualitative information from open-ended questions.

3.1.1 Results. A total of 582 logbooks have been collected between January 2022 and June 2022. The questionnaires have been analyzed from a descriptive statistical point of view regarding quantitative data while qualitative data have been investigated through thematic coding. From the aforementioned analysis emerged that more than half of the logbooks (52%) have been filled out by Greek teachers resulting in a possible polarization of the study toward their point of view. Other countries have answered respectively to 20% of logbooks, Bulgaria 17%, Romania and 11% Italy. The following quantitative data analyzed served the purpose to have a clear picture of the settings of the AI lectures so as to identify points of failure related to it and possible improvements in the development of future implementation. From the data emerged that the average reported class size was 16 students (SD= 6.9). The most frequent modality of teaching has been in-presence (N=503, 86%), followed by remote (N=57, 10%) and mixed modality (N=18, 3%). 197 lessons (34%) were related to Module 1, 134 (23%) to Module 2, 136 (23%) to Module 3, and 115 (20%) to Module 4. When asked about the material used in the lecture, the majority of respondents indicated

the use of computers (N=420) followed by video/audio recordings (N=306) and photo/images (N=285). Teachers were also asked to provide feedback on the perceived level of engagement of their pupils during the lectures. Students' engagement in each module has been rated between high, average and low. Module 2 and 4 were rated with "high" engagement 79% and 73% of the time respectively. The lower levels of engagement were reported for Modules 1 (high 63%, average 36%, low 1%) and 3 (high 61%, average 35%, low 4%); this could be related to the novelty of the topics for Module 1 or the complexity of the topics discussed for Module 3. The teaching material and the lesson plans provided to the teachers have been considered "adequate" for 87% of the lectures, "partially adequate" in 12% of the cases, and "inadequate" for the remaining 1%. The analysis of open-ended questions regarding "inadequate" and "partially adequate" responses underlined the following problems: (1) Need for more flexible materials (N=10), teachers reported not being able to make students perform specific activities because of the lack of technology; (2) Difficult content (N=7), some of the materials have been identified as too complex for the age range or level of knowledge of the students involved; (3) Difficult language (N=5), students had difficulty in comprehending the specific terms and the language used.

3.2 Focus group

The international focus group has been developed for gathering qualitative information regarding positive and negative experiences related to the implementation of the AI literacy program. The focus group was held online and involved 12 educators, with three educators from each country. It lasted one hour and eighteen minutes and comprised two main parts: (1) a personal experience description and (2) a discussion on the common points that emerged. During phase (1), representatives from each country shared their personal experiences and identified positive and negative aspects of the program and materials provided. Following this, the facilitator introduced phase (2) and used some of the critical aspects that emerged during the focus group as discussion points.

3.2.1 Results. The preliminary results from the focus group were analyzed through thematic analysis and they are summarized into six themes: (1) Time; (2) Engagement; (3) Technological Equipment; (4) Diversity; (5) Inclusion and (6) Best Practices.

Time. Time has been the most frequently reported limitation (N=11): participants found difficulties in finding time during the formal educational hours to conduct the lectures and devoted a considerable amount of time to understanding and adapting the material.

Engagement. Participants confirmed the high level of engagement of the pupils: the topic is perceived as interesting and relevant by the students. Moreover, the use of hands-on and project-based activities has been perceived positively by both students and educators. Teachers reported even higher levels of interest when students could clearly identify a link between AI applications and their personal life. For instance, students were very interested when discussing AI bias in facial recognition systems and the effects of this phenomenon in their daily lives. This was particularly true in highly multicultural contexts. In general, a good level of engagement has been shown by students to the ICT topics as a whole but,

in particular, greater interest has been noticed when coming to more practical activities. The hands-on activities included group work both requiring using digital tools or not. As reported also in the literature, physical activities and a mix of digital and physical ones positively impact AI learning experiences [18] and group activities on AI topics can facilitate the processing of information [7].

Technological Equipment. Old equipment and the absence of technological devices were also reported as limitations. Even if many activities included in the program did not require the use of technological devices (i.e., "unplugged" activities), in others the use of a computer was highly suggested and in some cases mandatory. Therefore the absence of devices, the poor performance of the old equipment, or the lack of internet connection hindered the program implementation.

Diversity. Diversity addressed by teachers has been mainly with respect to three dimensions: (1) digital literacy, (2) language, and (3) cultural and societal context. The digital literacy gap between students emerged in class while performing the activities and consisted of differences in the level of skills required to use the technology. Due to those differences, teachers had to rethink their lectures and slow down the pace of the program. Language has been another potential barrier when talking about AI literacy, especially in classes having a high percentage of students from other countries it has been really hard to find a common ground, linguistically speaking, to talk about those topics both for their novelty and complexity of them. The lectures related to ethics have been particularly difficult for the students because of their abstract nature and difficulty in expressing complex concepts using simple words. From the teacher's perspective, different cultural and societal contexts made it challenging to discuss with the whole class more theoretical concepts (such as the concept of algorithms or the ethical aspects) since students were not interested or motivated to study such topics.

Inclusion. The theme of inclusion was discussed using three different perspectives: (1) AI literacy to raise awareness on inclusion; (2) Inclusion as practice; (3) Inclusion reached through technology. Some topics covered in the program allowed the teachers to discuss with the class about inclusion: for instance, when exploring the concepts of bias in AI-powered facial recognition, the class discussed how inclusion and exclusion are intrinsic to AI systems as human-designed software. Teachers underlined the importance of inclusive practices as the default approach and not as optional activities. The use of tablets as supporting devices for some activities allowed all students to follow the lecture more effectively thanks to the real-time translations provided by the tool. Moreover, teachers reported that in some cases practical activities using digital devices fostered collaboration and inclusion of those pupils usually considered outsiders.

Best Practices. During the focus group, participants made suggestions and described best practices learned while bringing the activities into their classes. Those recommendations regarded (1) how to overcome the lack of time for preparing the activities; and (2) how to mitigate the complexity of the topic. Teachers suggested providing preparatory materials to the students before the classes to optimize the content to be covered during school hours. Another proposal to better manage the timing of the activities was to make AI literacy programs run through the whole year as curricular

subjects. Suggestions regarding how to mitigate complexity have been to provide ready-to-use examples and simplify the language. Providing a library of ready-to-use contextualized examples that present different AI-based technology and their main characteristics can support educators in adapting the examples according to their needs. Using a simple language can in general help to make AI literacy more accessible both to students with low digital literacy and also to pupils with low levels of understanding of the language of the country they live in.

4 DISCUSSION

The current study explored topics and best practices for inclusive AI literacy considering a teacher-centered approach. Three main topics can be distinguished by the preliminary analysis of the results: (1) AI Ethics, difficulties and opportunities; (2) Digital Competencies for AI Education; (3) Inclusive practices in AI education.

4.1 AI Ethics, difficulties and opportunities

The literature underlines the importance of addressing the ethical implications of AI when developing AI Education programs [12, 17, 25, 30]. While professionals and academics can be more informed about the positive and negative outcomes of the use of AI, the younger population can be misled by the representation of AI depicted by the media [17]. To allow younger generations to form a personal and critical opinion on the matter it is important not only to provide them with basic information about AI but also to foster ethical discussion about it too. For this reason, a specific module of the AI curriculum has been dedicated to the topic of AI and Ethics (Module 3). However, according to the data gathered from the research activities, this module was the one rated as the most difficult to discuss in class. Educators found the ethical implications of AI difficult to explain to their audience given the complexity of the topic itself, the implicitly abstract language, and the use of materials that might be difficult to understand by non-experts. To deal with those difficulties, some teachers devoted additional time to discussing the topic in class, simplifying the language used and presenting practical examples and scenarios that helped to make the topic closer to students' experience.

As suggested by Long and colleagues a good practice could be of framing ethical considerations using well-known tools as the centre of reflection [18]. For example, considering an audience of middle school students, YouTube could represent a good real-life example for discussing recommendation systems and their ethical challenges. Another approach could be the use of a game-based approach to surface ethical concerns [3].

Teachers underlined that once the difficulties regarding how to talk about ethics have been overcome, the students were engaged in positive and meaningful discussions. Reflecting upon possible bias introduced by developers of AI and perpetuated by the usage of such AI systems in the real world, can make students more aware of the role of AI as an extension of humans' capabilities and not as a standalone entity. Moreover talking about those topics fostered parallel discussions about diversity and inclusion. The use of examples closer to students' interests made them passionate about analyzing the ethical implications of the technological solutions proposed and made them aware of the human role in AI bias.

4.2 Digital Competences for AI Literacy

Digital Competencies represent a key aspect of current and future generations' life [9]. AI literacy requires indeed a basic level of digital literacy while scientific and computational literacy can additionally inform literacy on Artificial Intelligence even though they are prerequisite [17].

However, the disparity of digital competencies between students created difficulties for teachers to adapt the lectures to the whole class and implied additional time to assess the level of the pupils and create tailored materials based on their skills.

As a positive aspect, teachers' digital competencies developed along with students' competencies. In fact, AI literacy competencies are fundamental in the life of every citizen [4]. Considering that teachers involved in the project belonged to different subjects, having to teach AI topics improved also their personal knowledge of the topics and increased their digital competencies.

4.3 Inclusive practices in AI education

Inclusive education is considerably important also when talking about AI education programs [27]. For the sake of the current study, we considered inclusive education both as a form of education that brings equity among students and as an education that is accessible to the highest number of schools and students.

During the current investigation, a recurrent theme has been the power of interactive and practical activities performed during the initiative. Both plugged and unplugged activities promoted the engagement of pupils and allowed even students usually excluded to be actively involved in the class. This observation is in line with other studies showing the importance of laboratory and collaborative activities in AI education to increase engagement and improve learning in the topic [7, 16, 18].

Diversity might provide both opportunities and challenges given the multiplicity of needs at stake [22]. Learning activities therefore should be customized to students' needs and preferences and the use of technology should be weighted [22]. The findings of the current study align with previous research, which emphasized the need to customize AI programs not only at the class level but also at the country level [23]. The study's research activities revealed that some of the activities in the curriculum were adapted by teachers to reflect the country's cultural background. The role of technology in the AI education program was intentionally made optional, considering that the schools involved were located in disadvantaged areas and might not have continuous access to technology during class activities. For most activities, a non-digital version was provided. However, some difficulties arose during the program's implementation, as outdated technological equipment hindered the lectures' progress, resulting in missed opportunities for students. Other researchers, having observed the same issue, suggest the use of low-cost technology [22] or the use of mobile technology [14]. As observed in our study, the use of technological devices such as tablets could increase inclusion, whereas their absence or malfunction could decrease the participation of all students.

5 CONCLUSIONS

Synthesizing from the points stated above we summarise a preliminary list of best practices to develop an inclusive AI curriculum for

middle school.

Customization. Diversities between countries, schools, and individuals make it necessary to apply a customized approach to AI literacy. The first and foremost element of customization should be the language of the materials followed by every other aspect as significant examples based on the sociocultural setting, specific activities to target age differences, and activities flexibility based on the time and devices available.

Simplicity. The complexity of the AI topic paired with the lack of time for teachers requires a simplified approach to the subject. Nonetheless, the materials for students should be kept simple to respond to different language and comprehension levels.

Interactivity. Practical activities and interactivity allow a wider range of students to actively participate and be engaged in the lectures. Those activities also enhance collaboration between students resulting in a more inclusive learning environment.

Proper time and resources. Dedicated hours and adequate digital materials are fundamental for developing inclusive AI curricula. Lack of time implies - in most cases - leaving some students behind because of their difficulties. The absence of technological devices or their malfunctioning influence as well as AI literacy initiatives are those great tools for inclusion.

This study represents a starting point for realizing more inclusive AI literacy programs, considering the educators' perspective as central to making the next generation of AI literacy programs more inclusive and human-centered. A general limitation of this work is that the study involved different situations representing different educational contexts, school systems, and cultural settings rather than a uniform group.

Focus group voluntary participation could have possibly influenced the results considering the presumed positive attitude towards the topic. Being the language of the meeting was English in case of low proficiency some participants could have limited their interventions. Additionally, the presence of only a few representatives per country has possibly restricted the variety of information even more.

5.1 Future works

The current inquiry represents the first step toward more inclusive AI literacy programs in formal K-12 educational settings. Its exploratory nature requires further studies to transform its insight into concrete tools for AI literacy widespread.

The human-centred approach applied in this study considered the teachers as the main actor when it comes to education settings [10]. However further studies including the point of view of students will certainly contribute to creating a more complete picture of this emerging field [29].

AI education represents an innovative research trend because of the widespread of this emerging technology and its growth throughout the years [24]. The young generation has been born using all kinds of technological devices but has often not been educated for a critical use [5]. For this reason, it is important to offer youths the possibility to develop proper knowledge about the major technological trends from a practical, theoretical and reflective perspective. Creating inclusive AI literacy programs for K-12 represents both a great opportunity and challenge that, if faced

on time, could ensure a brighter future in which AI is not blindly feared but understood and exploited in humans' best interest.

ACKNOWLEDGMENTS

The authors would like to thank all the educators involved in the study. This research was supported by the Erasmus+ program – project DIG4Future, Digital competencies, Inclusion and Growth for Future generations (project ID 400621507)

REFERENCES

- [1] Mel Ainscow, Jean Conteh, Alan Dyson, and Frances Gallanagh. 2007. *Children in primary education: demography, culture, diversity and inclusion*. Primary Review, Cambridge. OCLC: 276484994.
- [2] Hadi Ali and Andrew David Maynard. 2021. Design the Future Activities (DFA): A Pedagogical Content Knowledge Framework in Engineering Design Education. *ASEE PEER*. <https://doi.org/10.18260/1-2--36924>
- [3] Stephanie Ballard, Karen M. Chappell, and Kristen Kennedy. 2019. Judgment Call the Game: Using Value Sensitive Design and Design Fiction to Surface Ethical Concerns Related to Technology. In *Proceedings of the 2019 on Designing Interactive Systems Conference*. ACM, San Diego CA USA, 421–433. <https://doi.org/10.1145/3322276.3323697>
- [4] Stephanie Carretero, Riina Vuorikari, Yves Punie, and European Commission. Joint Research Centre. 2017. *DigComp 2.1 : the digital competence framework for citizens with eight proficiency levels and examples of use*. Publications Office.
- [5] European Commission. 2022. *Digital Economy and Society Index (DESI)*. Technical Report. European Commission. <https://digital-strategy.ec.europa.eu/en/policies/desi>
- [6] Stefania Druga, Fee Lia Christoph, and Amy J Ko. 2022. Family as a Third Space for AI Literacies: How do children and parents learn about AI together?. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, NY, USA, 1–17. <https://doi.org/10.1145/3491102.3502031>
- [7] Stefania Druga and Amy J Ko. 2021. How do children's perceptions of machine intelligence change when training and coding smart programs?. In *Interaction Design and Children*. ACM, Athens Greece, 49–61. <https://doi.org/10.1145/3459990.3460712>
- [8] Amy Eguchi, Hiroyuki Okada, and Yumiko Muto. 2021. Contextualizing AI Education for K-12 Students to Enhance Their Learning of AI Literacy Through Culturally Responsive Approaches. *KI - Künstliche Intelligenz* 35, 2 (2021), 153–161. <https://doi.org/10.1007/s13218-021-00737-3>
- [9] European Commission. Directorate General for Education, Youth, Sport and Culture. 2019. *Key competences for lifelong learning*. Publications Office, LU. <https://data.europa.eu/doi/10.2766/291008>
- [10] Joanna Goode. 2007. If You Build Teachers, Will Students Come? The Role of Teachers in Broadening Computer Science Learning for Urban Youth. *Journal of Educational Computing Research* 36, 1 (Jan. 2007), 65–88. <https://doi.org/10.2190/2102-5G77-QL77-5506>
- [11] Maya Israel, Supriya Dey, Yota Dimitriadis, Heather Feldner, Amy Isvik, Noel Kuriakos, Jean Salac, Mina Huh, and Gesu India. 2021. Reimagining Accessibility and Inclusion in K-12 CS Education through curriculum and professional development. <https://www.microsoft.com/en-us/research/uploads/prod/2021/02/Reimagining-Acc-and-Inclusion-in-K12-Making-Higher-Ed-Group-B.pdf>
- [12] Yasmin Kafai, Chris Proctor, and Debora Lui. 2019. From Theory Bias to Theory Dialogue: Embracing Cognitive, Situated, and Critical Framings of Computational Thinking in K-12 CS Education. In *Proceedings of the 2019 ACM Conference on International Computing Education Research*. ACM, Toronto ON Canada, 101–109. <https://doi.org/10.1145/3291279.3339400>
- [13] Siu Cheung Kong, William Man-Yin Cheung, and Olson Tsang. 2022. Evaluating an artificial intelligence literacy programme for empowering and developing concepts, literacy and ethical awareness in senior secondary students. *Education and Information Technologies* (Oct. 2022). <https://doi.org/10.1007/s10639-022-11408-7> MAG ID: 4307157265.
- [14] Chrysoula Lazou and Maria Psychogiou. 2020. Mobile Devices as a Vehicle for Inclusive Educational Practices: A Case Study. *Journal of Modern Education Review* 10 (2020).
- [15] Phoebe Lin and Jessica Van Brummelen. 2021. Engaging Teachers to Co-Design Integrated AI Curriculum for K-12 Classrooms. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–12. <https://doi.org/10.1145/3411764.3445377>
- [16] Duri Long, Takeria Blunt, and Brian Magerko. 2021. Co-Designing AI Literacy Exhibits for Informal Learning Spaces. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (Oct. 2021), 1–35. <https://doi.org/10.1145/3476034>
- [17] Duri Long and Brian Magerko. 2020. What is AI Literacy? Competencies and Design Considerations. *Proceedings of the 2020 CHI Conference on Human Factors*

- in *Computing Systems* (2020). <https://doi.org/10.1145/3313831.3376727>
- [18] Duri Long, Aadarsh Padiyath, Anthony Teachey, and Brian Magerko. 2021. The Role of Collaboration, Creativity, and Embodiment in AI Learning Experiences. In *Creativity and Cognition*. ACM, Virtual Event Italy, 1–10. <https://doi.org/10.1145/3450741.3465264>
 - [19] Duri Long, Anthony Teachey, and Brian Magerko. 2022. Family Learning Talk in AI Literacy Learning Activities. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, NY, USA, 1–20. <https://doi.org/10.1145/3491102.3502091>
 - [20] Davy Tsz Kit Ng, Jac Ka Lok Leung, Samuel Kai Wah Chu, and Maggie Shen Qiao. 2021. Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence* 2 (Jan. 2021), 100041. <https://doi.org/10.1016/j.caeai.2021.100041>
 - [21] Sara Polak, Gianluca Schiavo, and Massimo Zancanaro. 2022. Teachers' Perspective on Artificial Intelligence Education: an Initial Investigation. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. ACM, New Orleans LA USA, 1–7. <https://doi.org/10.1145/3491101.3519866>
 - [22] Sdenka Zobeida Salas-Pilco, Kejiang Xiao, and Jun Oshima. 2022. Artificial Intelligence and New Technologies in Inclusive Education for Minority Students: A Systematic Review. *Sustainability* 14, 20 (Oct. 2022), 13572. <https://doi.org/10.3390/su142013572>
 - [23] Jiahong Su, Yuchun Zhong, and Davy Tsz Kit Ng. 2022. A meta-review of literature on educational approaches for teaching AI at the K-12 levels in the Asia-Pacific region. *Computers and Education: Artificial Intelligence* 3 (2022), 100065. <https://doi.org/10.1016/j.caeai.2022.100065>
 - [24] Bergur Thormundsson. 2022. Global AI software market growth 2019-2025. <https://www.statista.com/statistics/607960/worldwide-artificial-intelligence-market-growth/#statisticContainer>
 - [25] David Touretzky, Christina Gardner-McCune, Fred Martin, and Deborah See-horn. 2019. Envisioning AI for K-12: What Should Every Child Know about AI? *Proceedings of the AAAI Conference on Artificial Intelligence* 33 (2019), 9795–9799. <https://doi.org/10.1609/aaai.v33i01.33019795>
 - [26] UNESCO. 2022. UNESCO World Report: Investing in Cultural Diversity and Intercultural Dialogue. https://www.un.org/en/events/culturaldiversityday/pdf/Investing_in_cultural_diversity.pdf
 - [27] UNICEF. 2014. *Conceptualizing Inclusive Education and Contextualizing it within the UNICEF Mission*. Technical Report. <https://www.unicef.org/lac/media/35101/file/Ingl%C3%A9s.pdf>
 - [28] UNICEF. 2019. Digital literacy for children – 10 things to know. <https://www.unicef.org/globalinsight/documents/digital-literacy-children-10-things-know>
 - [29] Ge Wang, Jun Zhao, Max Van Kleek, and Nigel Shadbolt. 2022. 'Don't make assumptions about me!': Understanding Children's Perception of Datafication Online. *Proc. ACM Hum.-Comput. Interact* 6, 419 (Nov. 2022). <https://doi.org/10.1145/3555144>
 - [30] Randi Williams, Safinah Ali, Nisha Devasia, Daniella DiPaola, Jenna Hong, Stephen P. Kaputsos, Brian Jordan, and Cynthia Breazeal. 2022. AI + Ethics Curricula for Middle School Youth: Lessons Learned from Three Project-Based Curricula. *International Journal of Artificial Intelligence in Education* (Aug. 2022). <https://doi.org/10.1007/s40593-022-00298-y>
 - [31] Olaf Zawacki-Richter, Victoria I. Marin, Melissa Bond, and Franziska Gouverneur. 2019. Systematic review of research on artificial intelligence applications in higher education – where are the educators? *International Journal of Educational Technology in Higher Education* 16, 1 (Dec. 2019), 39. <https://doi.org/10.1186/s41239-019-0171-0>
 - [32] Michelle Renée Zimmerman. 2018. *Teaching AI: exploring new frontiers for learning*. International Society for Technology in Education, Portland, Oregon.